Team meeting — 11 November 2026 (class group time)

Mocap3D (2D video → 3D) · What to discuss and who does what next · Background: *Follow-up Experiments from the Paper Review* (Paper_Followup_Experiments_EN.pdf)

> **Goal of this meeting**: share the three follow-up results, make three decisions, and leave with one clear task per person.

1. Agenda (about 40 minutes)

Time	Item	Lead
5 min	Thank you for the paper review. Map each paper to the experiment it led to (report, section 1).	Sonny
10 min	Three results, one minute each (section 2 below)	Sonny
15 min	Three decisions (section 3)	All
10 min	Roles and due dates (section 4); questions	All

2. What to tell the team (one sentence each)

1. **VideoPose3D idea → confirmed.** Our 81-frame window is what makes the skeleton smooth. Without it, the velocity error nearly triples (7.0 → 19.5 mm/frame). Our output is smoother than Kinect's (7.0 vs 11.5). A simple 0.3 s smoothing filter makes it even smoother, at no cost.
2. **REHAB24-6 idea → depth is our weak point.** Our largest error is along the depth axis (38 mm vs 23–26 mm across the image). For Kinect, a depth camera, depth is the best axis. Torso rotation (RTK) has the most depth error. The exercise matters more than the camera view (REHAB24 Ex5: 112 mm).
3. **3D feedback idea → only partly works.** A training-free DTW comparison with correct repetitions flags wrong CTK and RTK repetitions about as well as Kinect (AUC 0.73 / 0.70). ELK fails for every source. Naming the wrong joint agrees with Vicon only 22–35 % of the time, so we should **not** claim joint-level feedback yet.

All three: only 3 KERAAL G3 participants (healthy, acted errors). Intervals overlap between sources. KERAAL is CC BY-NC-SA: never share raw data or per-frame outputs publicly.

3. Decisions to make together

#	Question	Proposal (Sonny)	Decision
D1	Add the smoothing filter to the final pipeline?	Yes. A 0.3 s Hamming filter after lifting, reported as a separate step.	
D2	What do we claim about feedback in the final presentation or paper?	Repetition-level error detection ("this repetition looks wrong"), not joint-level ("your left knee is wrong"). Joint-level stays as future work.	

D3	Do we request the UCO Physical Rehabilitation dataset (Research 1)?	Yes, request now; access may take time. Use it only if it arrives in time, as an extra test of cameras and exercises.	

4. Proposed roles (to confirm in the meeting)

Person	Task	What to deliver	Due
Sonny	Depth on rotation (RTK): try more rotation examples in training, or VideoPose3D back-projection on 1A videos. Add the smoothing filter (D1) to the demo pipeline.	RTK depth error before/after; updated demo video	
Roshini (programming)	Re-run the supervised error classifier (RQ4, random forest) on the newest model's 3D. Make one figure that shows *when* a repetition differs from a correct reference, using the DTW alignment.	AUC table (ours / Kinect / Vicon); one timing figure	
Euijin (data)	(a) Find the REHAB24-6 names of Ex1–Ex6 and why Ex5 is hard. (b) Check what the KERAAL error codes (E1–E3, B1–B3) mean for ELK, to explain why ELK fails. (c) Send the UCO dataset request if D3 = yes.	Short table of exercise names and error codes; request sent	
Joshi	Read the follow-up report and mark any sentence or term that is unclear. These become the plain-language glossary for the presentation.	List of unclear points (optional, no coding)	

The roles follow the earlier team split: Euijin = data, Roshini = programming, Joshi = understanding the project, no heavy tasks. Fill in the due dates together.

5. Questions to ask each person

- **Euijin**: what did you find when you compared Kinect-like 3D tools with Mocap3D? Did `pr11_results.zip` and `share_merge_1106.zip` raise any problems? Those files predate the REHAB24 model.
- **Roshini**: is anything blocked (GPU, data access, PR #10/#11)?
- **Everyone**: which of the four papers should we cite as our main method source? Proposal: VideoPose3D for the model, REHAB24-6 for the evaluation idea.

Notes